

Ethan Zhang

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Education

University of Illinois Urbana-Champaign

May 2026

B.S. in Computer Engineering | GPA: 3.7/4.0

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Data Mining, AI, Cloud Computing

Experience

PTC

May 2025 – Aug 2025

Software Development Intern

- Deployed filtering and name-based search by implementing Elasticsearch index integration, Angular component architecture, and Bootstrap scalable layouts, achieving a **400%** increase in filter options and a **15-second** reduction per MongoDB query
- Enhanced project management system by refactoring REST API endpoints and developing Angular and Vue.js frontend components, introducing multi-assignee project management workflows for over **4 million+** active users
- Strengthened CI/CD pipelines by creating numerous end-to-end Playwright tests, which expanded the internal project management systems' front-end automated test coverage by **33%**

IPG Photonics

May 2024 – Aug 2024

Software Automation Intern

- Increased camera station production output by **25%** and avoided **\$6,400** in licensing costs by migrating camera-vision station archetypes from NI Vision to OpenCVSharp
- Delivered instant spec reporting for production teams that cut cycle times from **10 minutes** to **5 seconds** by building a full-stack .NET Core application with SQL and Entity Framework Core for automated report generation
- Reduced operator errors by **~15%** and improved traceability across **1000+** stations in **25** different station archetypes by refactoring the universal identification system in the common base archetype

University of Illinois Urbana-Champaign

May 2021 – Aug 2021

Research Assistant

- Designed Unity (OpenXR) VR medical labs with a data-driven workflow engine and world-space UI/voice prompts, delivering step-by-step procedure exposure to **40+** students
- Implemented xAPI-style events (step entered/completed, retries, time-on-step) and integrated Cognitive3D for interaction analytics to track and display key metrics in Metabase dashboards
- Improved time-to-publish for new VR scenarios by implementing a modular workflow framework in Unity (C#, ScriptableObjects, Addressables) that lets students configure all stages of medical scenarios without developer support

Projects

Quantum Match

- Developed facial recognition kiosk museum exhibit prototype for the **\$500M** Illinois Quantum & Microelectronics Park, implementing a full matching pipeline with OpenCV and InsightFace to run against pre-computed facial embeddings within a **4GB RAM** constraint
- Built the data acquisition layer for the exhibit by developing a production Angular/TypeScript portal with multi-step validation, cloud file uploads, and Supabase persistence to collect professor profiles and multimedia assets fed into the OpenCV matching pipeline

NFL Fantasy Breakout Predictor

- Predicted weekly breakout NFL candidates by training a gradient-boosted tree model (XGBoost) that achieved a **16%** higher breakout hit rate over ESPN waiver-wire recommendations for players rostered in under 25% of all 2 million active leagues
- Drove an average **7%** week-over-week increase in model accuracy by organizing position-specific datasets from nflreadpy, nfl-data-py, and web-scrappers into a calculated weekly player table for continuous retraining and fine-tuning

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C#, HTML/CSS, C++, SQL, Go

Frameworks: Angular, .NET, React, Nest.js, Spring Boot, PyTorch, TensorFlow, Scikit-learn, Node.js, Streamlit, Vue.js

Developer Tools: AWS, Docker, Kibana, Playwright, Nginx, Elasticsearch, RabbitMQ